

COS-AT1 - Pseudocode Writing Task

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Kahn's algorithm for resource allocation:

Modify Kahn's algorithm so that it only processes a node if the "required allocation" (a property of the node) are less than "Availability system memory". update the memory as nodes are processed.

ALGORITHM :

1. calculate the in-degree of Every node.
2. Find nodes with in-degree 0.
3. Process a node only if its required resources are less than or equal to available system memory.
4. update available memory after processing the node.
5. Decrease the in-degree of its adjacent nodes.
6. Repeat until all possible nodes are processed.
7. If all nodes are processed, return the topological order; otherwise, report that ordering is not possible due to resource constraints.

PSEUDO CODE :

ALGORITHM Resource Based Kahn (G , Available Memory)

BEGIN

FOR each vertex v in G DO

InDegree [v] $\leftarrow 0$

END FOR

FOR each edge (u, v) in G DO

InDegree [v] $\leftarrow$ InDegree [v] + 1

END FOR

$Q \leftarrow$ empty Queue

Result $\leftarrow$ empty list

FOR each vertex v in G DO

IF InDegree [v] = 0 AND

Required Resource [v] $\leq$ Available Memory THEN

ENQUEUE (Q, v)

END IF

END FOR

WHILE Q is not empty DO

$u \leftarrow$ DEQUEUE (Q)

IF Required Resources [u] $\leq$ Available Memory

ADD u to Result

- THEN

FOR each neighbour v of u DO InDegree

[v] $\leftarrow$ InDegree [v]

IF InDegree [v] = 0 AND Required Resource

[v] $\leq$ Available Memory THEN

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    ENQUEUE (Q, v)
  END IF
END FOR

AvailableMemory ← AvailableMemory + RemainingResource

  [u]
END WHILE

IF length( Result ) = number of Vertices in G THEN
  PRINT " All nodes Processed "
ELSE
  PRINT " Some nodes can't be processed "
END IF
END

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d. Lexicographical Topological Sort :

Write Pseudocode for a Topological Sort that, when faced with multiple tasks, having zero in-degree, always picks the tasks whose name comes first alphabetically.

Idea : When several nodes have in-degree 0, always select the node whose name comes first alphabetically.

PSEUDO CODE :

LEXICOGRAPHICAL - TOPOLOGICAL - SORT (G)

for each node v in G

indegree [v] ← number of incoming edges

Q ← empty MIN-PRIORITY QUEUE

for each node v in G

if $\text{indegree}[v] = 0$

INSERT v INTO Q

result $\leftarrow$ empty list

while Q is not empty

$v \leftarrow \text{REMOVE-MIN}(Q)$

ADD v to result

for each neighbour u of v

$\text{indegree}[u] \leftarrow \text{indegree}[u] - 1$

if $\text{indegree}[u] = 0$

INSERT u into Q

if $\text{length}(\text{result}) = \text{number of nodes}$

return result

else

Print "cycle exists"

eg:

if the zero in-degree tasks are:

A, C, B

The algorithm always selects:

$A \rightarrow B \rightarrow C$

because alphabetical order is used whenever

there is a choice